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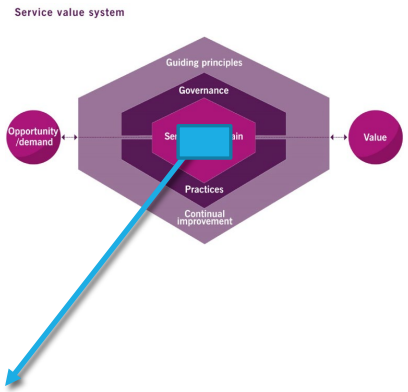
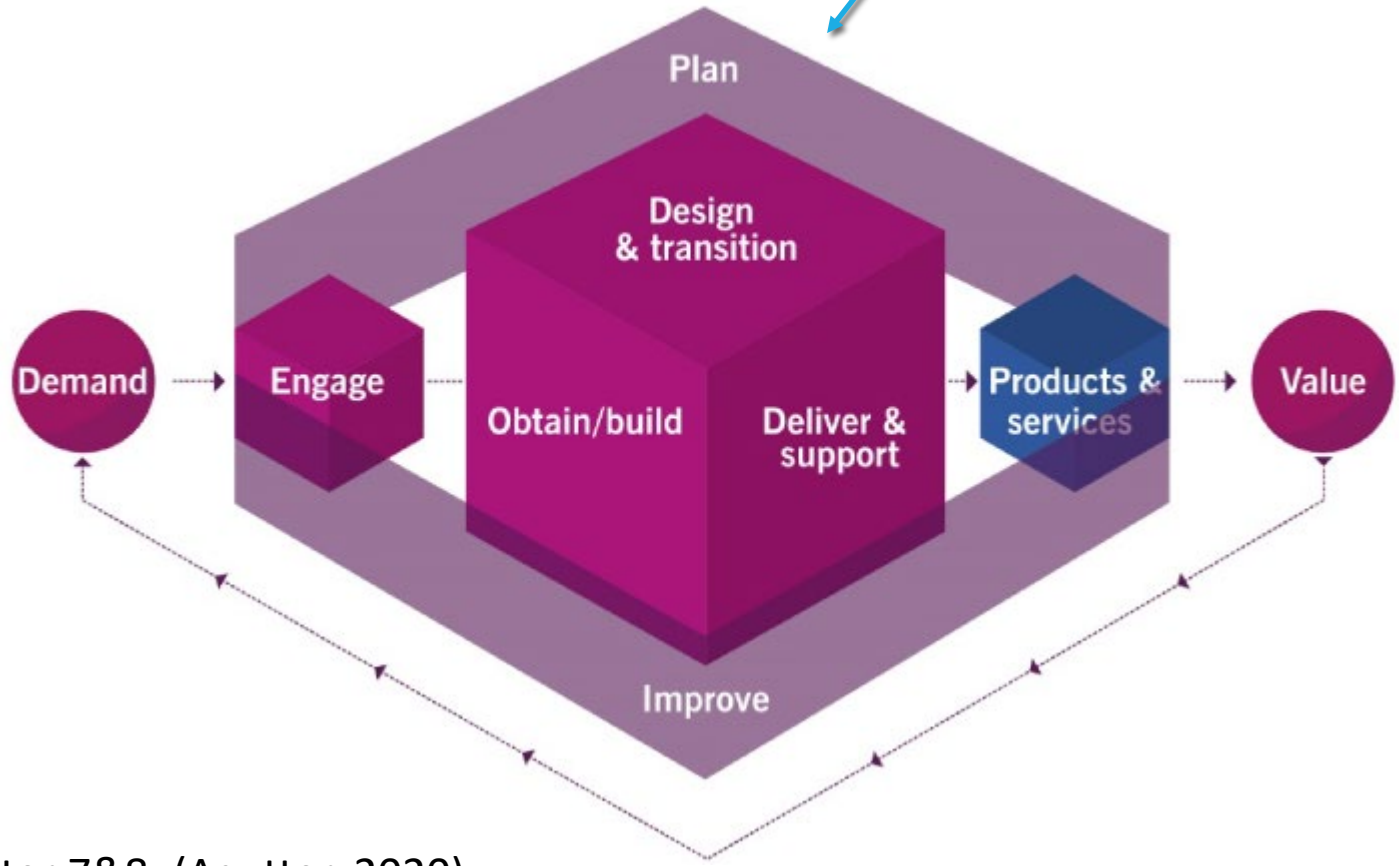
# IT Service Management (ITSM)

2025




## Service value chain




Chapter 7&8; (Agutter, 2020)

# Practices

The ITIL 4 Foundation syllabus requires you to know the purpose of 15 ITIL practices. For some of these 15, you also need to know key terms, and there are 7 practices that are studied in more depth.

| Practices |
| --- |
| **Information Security Management** |
| **Relationship Management** |
| **Supplier management** |
| **IT asset management** |
| **Monitoring and Event management** |
| **Release management** |
| **Service configuration management** |
| **Deployment management** |

# *Information security management*

*"The purpose of the information security management practice is to protect the information needed by the organization to conduct its business. This includes understanding and managing risks to the confidentiality, integrity, and availability of information, as well as other aspects of information security such as authentication (ensuring someone is who they claim to be) and non-repudiation (ensuring that someone can't deny that they took an action)."*

# *Relationship management*

*"The purpose of the relationship management practice is to establish and nurture the links between the organization and its stakeholders at strategic and tactical levels. It includes the identification, analysis, monitoring and continual improvement of relationships with and between stakeholders."*

# *Supplier management*

*"The purpose of the supplier management practice is to ensure that the organization's suppliers and their performances are managed appropriately to support the seamless provision of quality products and services. This includes creating closer, more collaborative relationships with key suppliers to uncover and realize new value and reduce the risk of failure."*

# *IT asset management*

*"The purpose of the IT asset management practice is to plan and manage the full lifecycle of all IT assets, to help the organization:*

- *Maximize value*
- *Control costs*
- *Manage risks*
- *Support decision making about purchase, re-use, retirement and disposal of assets*
- *Meet regulatory and contractual requirements."*

**An IT asset is *"any financially valuable component that can contribute to the delivery of an IT product or service."***

***Examples of IT-asset:*** *computers, servers, backup systems, switches, routers, software etc…*

# *Monitoring and Event management*

*"The purpose of the monitoring and event management practice is to systematically observe services and service components, and record and report selected changes of state identified as events. This practice identifies and prioritizes infrastructure, services, business processes, and information security events, and establishes the appropriate response to those events, including responding to conditions that could lead to potential faults or incidents."*

An **event** is *"any change of state that has significance for the management of a service or other configuration item. Events are typically recognized through notifications created by an IT service, configuration item or monitoring tool."*

# *Monitoring and* Event management

**Definition:** an event is *"**any change of state that has significance for the management of a service**. Events are typically recognized through notifications created by an IT service, configuration item or monitoring tool."*

Event management requires good **monitoring** tools and are classified in one of three ways:

- **Informational** – no action required, but may form part of a trend or data for analysis.
- **Warning** – action needs to be taken to avoid negative impact.
- **Exception** – action required; negative business impact might already have occurred.

# *Release management*

*"The purpose of the release management practice is to make new and changed services and features available for use."*

A release is *"a version of a service or other configuration item, or a collection of configuration items, that is made available for use."*

# Why Release Management?

- Ensures….
  - broader support for large scale transitions and releases
  - support for an organization anticipating unusual volume of change
  - That release packages are built and installed according to plan
  - Transition of knowledge to customers
  - That there is a release identification number

- …and improves Service Validating and Testing
  - Provides separate and more focused support for testing prior to release
  - Quality and error control are of paramount importance

- …and evaluation
  - Provides support for post release evaluation and confirmation of customer acceptance of new and changed services

# Different types of releases

- Small/minor release: A component or a minor reconfiguration.
- Emergency release: Changes that are assigned the highest priority (level 1).
- Large/major release: A whole system
- Bundled release: A group of software to be installed

- Discuss: Big bang release vs phased release
- Discuss: Pushed release vs pulled release

# Deployment management

*"The purpose of the deployment management practice is to move new or changed hardware, software, documentation, processes, or any other component to live environments. It may also be involved in deploying components to other environments for testing or staging."*

# Incident Management

*"The purpose of the incident management practice is to minimize the negative impact of incidents by restoring normal service operation as quickly as possible."*

An incident is *"an unplanned interruption to or reduction in the quality of a service"*.

A "Normal operation" is usually defined in an SLA

# Incident Management

- Incident:
  - Any occurrence which causes or may cause interruption or degradation to an IT service
  - Priority should reflect the impact

vs

- Problem / Error:
  - The unknown underlying cause of one or more incidents
  - Known Error
  - Known cause of incident for which workaround also exists

# Incident Management

- Some examples of what an incident may be are:
  - The fact that a user can not receive or send email
  - A computer cannot access the network
  - The fact that an application has been slower than usual concerning response times
  - The fact that an application is down / does not work at all
  - The “search feature” on the company website does not work.

An incident is often categorized in accordance to the name of the service. However, that depends on how the organization has administered their specific process.

# *Too: Swarming*

Some organisations use an incident management technique called swarming. A group of stakeholders work together until it becomes clear who is best placed to continue with the incident. Collaboration like this supports information sharing and provides learning opportunities within the teams.

# Incident Management - Priority

- Based on the criticality of the incident and how much impact it has on business a priority is assigned to incident.
- All incident cases are assigned a priority
- The scale that could be used is priority level 1 to 4. Level 1 is the highest priority and involves a critical incident.
- It is important that all staff understand the meaning of this priority scale (e.g., 1, 2, 3, and 4 respectively)

# Incident priority (example)

**Incident Priority 1:**

- Persistent and transient incidents that enables value is unavailable.
- Loss of a critical system or function.
- Loss of a critical digital service.

**Incident Priority 2:**

- Reduction of a critical system or a critical application (i.e., service)
- Loss of a non-critical application.

**Incident Priority 3:**

- Reduced functionality of a non-critical function.
- Reduced functionality of a non-critical application.
- Loss of a non-critical function.

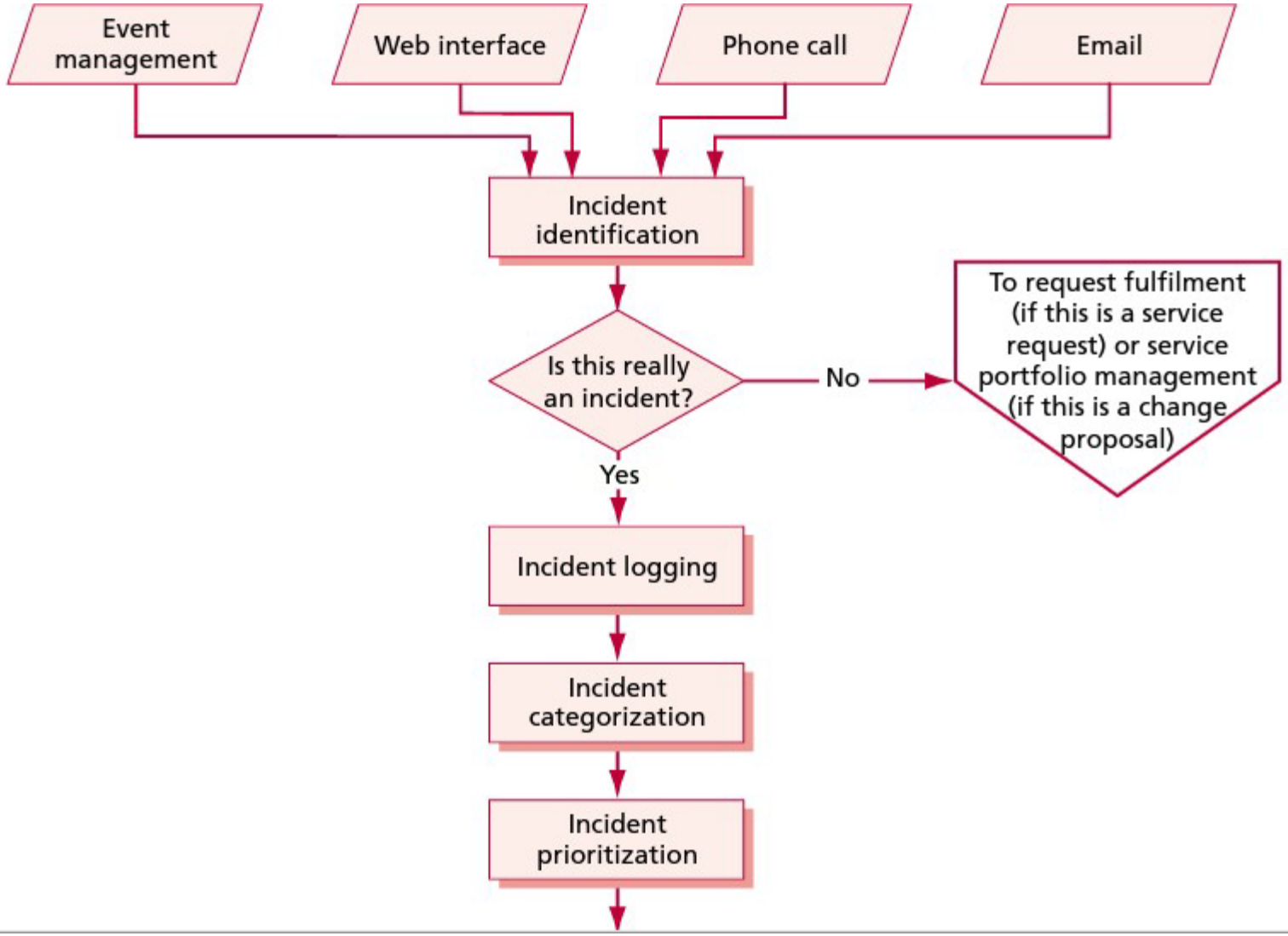
**Incident Priority 4:**

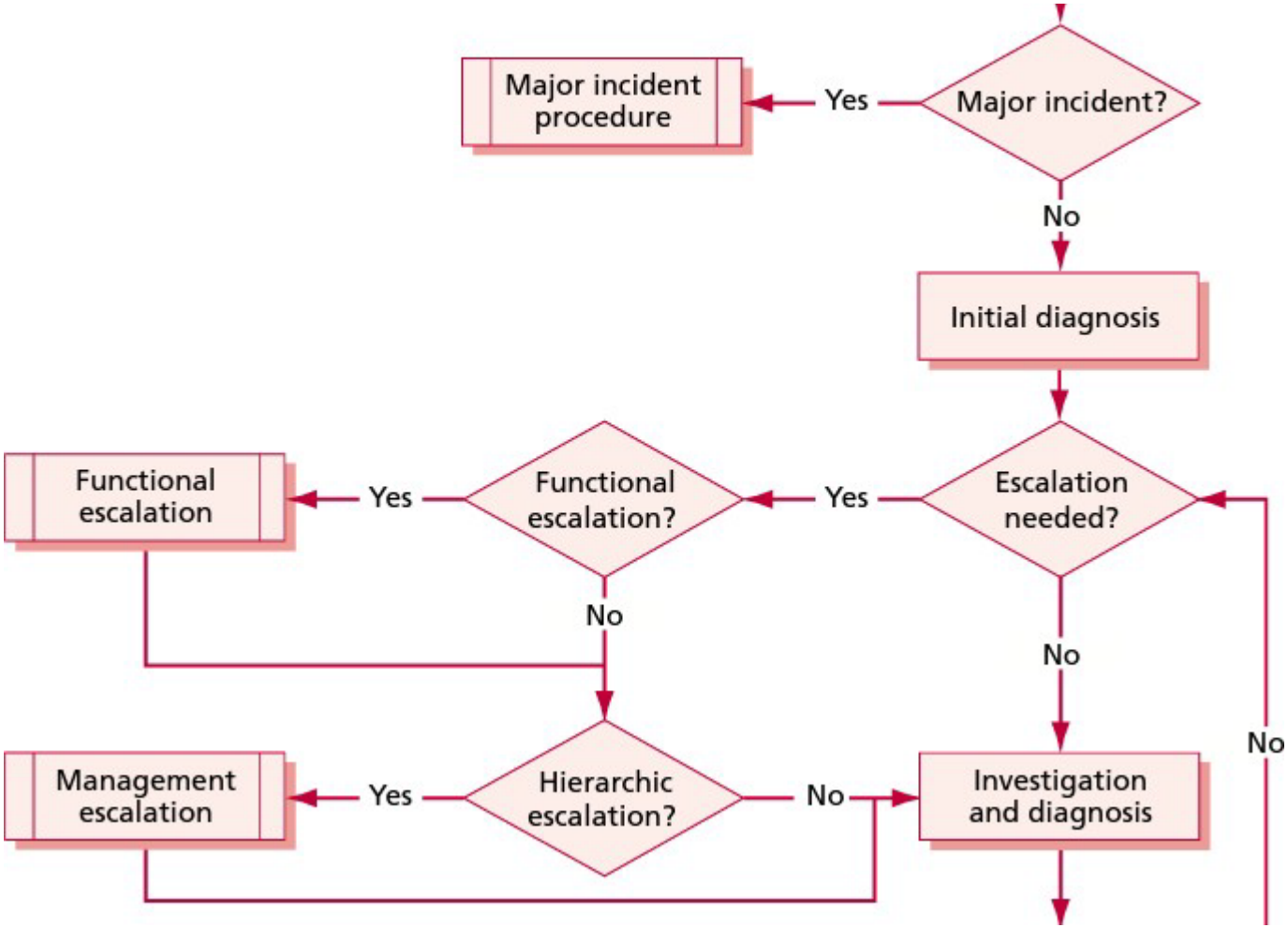
- The incident has little or no impact on the core business.
- Reduced functionality for non-critical IT services.

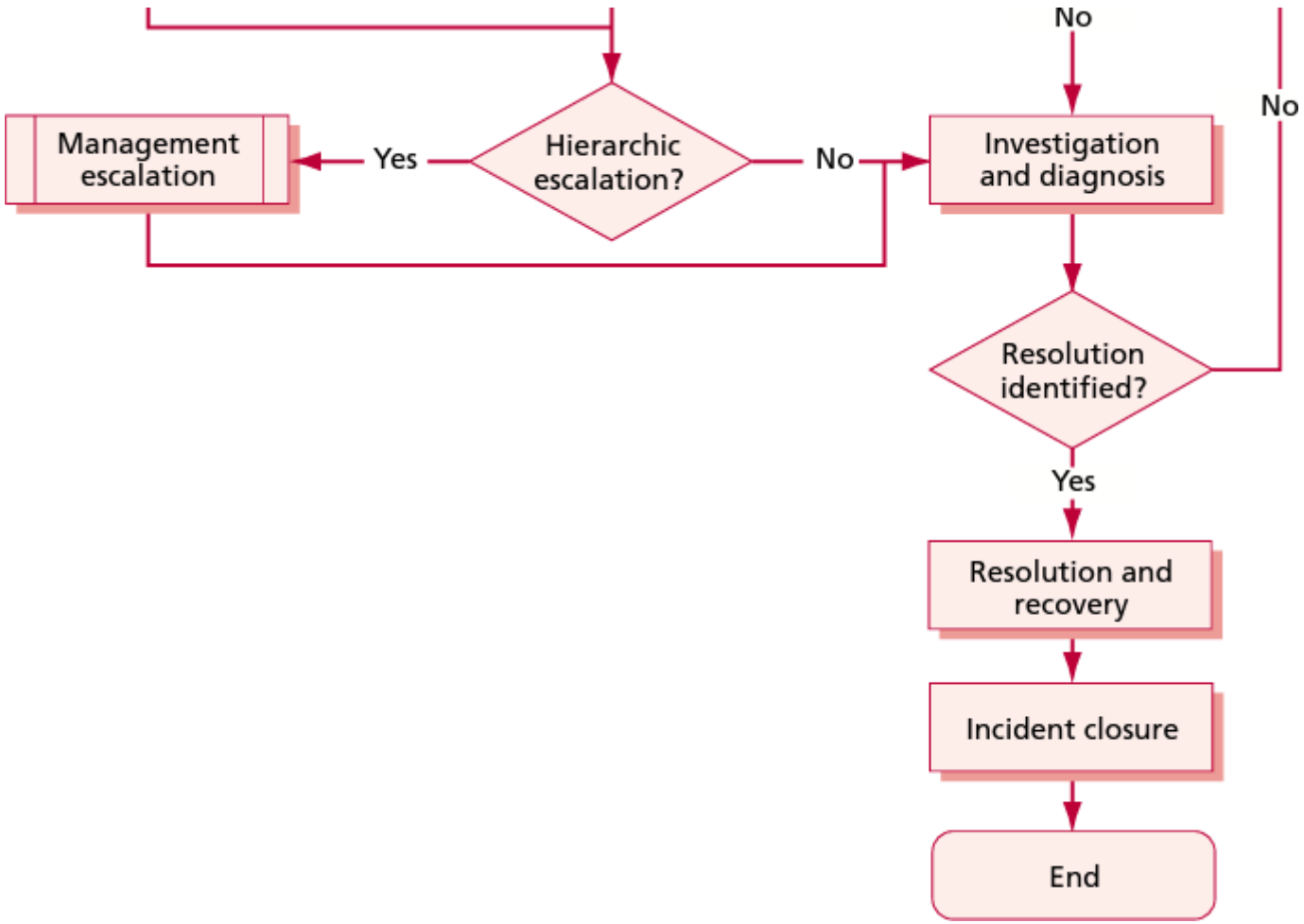
**incidents with the highest impact and urgency need to be resolved first.**
**Classifications and timescales are agreed with consumers.**

# NOTE! This is a simple example of incident mgmt process/practice






Major incident procedure
Yes
Major incident?
No
Initial diagnosis
Functional escalation
Yes
Functional escalation?
Yes
Escalation needed?
No
No
Management escalation
Yes
Hierarchic escalation?
No
Investigation and diagnosis
No


No
No
Management escalation
Yes
Hierarchic escalation?
No
Investigation and diagnosis
Resolution identified?
Yes
Resolution and recovery
Incident closure
End

# SO: Problem Management

*The purpose of the problem management practice is to reduce the likelihood and impact of* ***incidents by identifying actual and potential causes of incidents*** *and* ***managing workarounds and known errors***

***Definition****: A problem is a cause, or potential cause, of one or more incidents.*

Problems require investigation and analysis to identify the causes

*¨Agutter 2020*

# SO: Problem Management

- To identify the **root cause** of incidents
- Once the root cause is determined, Request For Change (RFC) is issued to initiate action towards implementation of a permanent fix
- **Workaround** is a solution that reduces or eliminates the impact of an incident or problem for which a full resolution is not yet available. Some workarounds reduce the likelihood of incidents. (Agutter 2020)
- Production and maintenance of Known Error Database (KEDB) is one of the most important outputs (part of knowledge mgmt. process/practice). **A known error is a problem that has been analyzed but not yet resolved** (Agutter 2020)

# SO: Problem Management

Problem management is **reactive, proactive and preventive**:

- **Reactive work** includes providing support to the incident management process, identify problems and solve problems that have caused one or more incidents. Usually driven by the incident management process
- **Proactive work** includes identifying and solving problems proactively by a service provider to reduce “fire fighting” activities. It also includes preventive work that is intended to solve errors in the IT service.
- **Preventive work** can involve regularly performed, planned tasks that are scheduled based on either time passed or meter triggers. Improvement.

# Service request management

*"The purpose of the service request management practice is to support the agreed quality of a service by handling all pre-defined, user-initiated service requests in an effective and user-friendly manner."*

A service request is *"a request from a user or a user's authorized representative that initiates a service action which has been agreed as a normal part of service delivery."*

# Service request management

- Service Request: a general term for all requirements and requests
- Does not include incidents or problems
- Service requests is usually owned by service desk
- Following a process aiming at assist users in situations where no service degradation or interruption is involved
- Frequently makes use of standard changes and automation to meet user requests
- Simple examples:
  - Request for information
  - Request for provision of a resource or service
  - Request for access to a resource or service
  - Feedback, compliments and complaints

# Change Management Practices

IThe purpose of the change enablement practice is to maximize the number of successful service changes by ensuring that risks have been properly assessed, authorizing changes to proceed, and managing the change schedule.

Change enablement must balance delivering benefits through successful changes and protecting live service from harmful changes.

# Definition

**Definition of change**: *A change is the addition, modification, or removal of anything that could have a direct or indirect effect on services.*

# Change Practice

- The practice supports an organization to:
  - Act on new customer requirements
  - Act on requirements regarding adaptation to the business and operation environment
  - Secure that the requirements is handled according to the actual rules (documentation, evaluation, testing etc.)
  - Secure that all modifications will be documented in CMS (Configuration Management System)
  - Optimize the business risks in connection to the change activities
- By establish a Change Advisory Board (CAB) useful people can be enrolled for support

# Change Advisory Board (CAB)

- A Change Advisory/Approval Board:
  - …is a group of experts convened by a Change Manager
  - …approves or reject change requests
  - …shall ensure that impacts are being investigated for how a change might have on the impact on IT infrastructure in terms of capacity, availability, continuity and security. All according to the service level agreements that may be signed by the customer.
  - …prioritize issues/errands

# Change Advisory Board (CAB)

An agenda for change meetings may include changes that have failed, change matters to be approved or rejected and audit of implemented changes. Change and control group handles old, new and future changes.

Examples of roles that can be represented in the change advisory board are:

- Change manager
- Representative of the client (can be customer manager)
- Development
- Infrastructure
- Problem manager
- Release manager
- External suppliers
- User support
- Service level manager

# Types of Change

| | |
|---|---|
| **Standard change** | Standard changes are low-risk, preauthorised changes. They are well understood and documented so they can be implemented without additional authorisation. An example of a standard change could be giving a new starter access to a piece of approved software. |
| **Normal change** | Normal changes need to be scheduled, assessed and authorised via the organisation's defined process. Lower-risk changes will need less scrutiny than high-risk changes. Many organisations have tools in place that manage change request workflows, automating the process where it makes sense to do so. |
| **Emergency change** | Emergency changes need to be implemented as soon as possible, perhaps in response to an issue or a security breach. They are assessed and authorised when possible, but some steps (e.g. testing) might be left out if the level of urgency justifies it. There may be a separate change authority for emergency changes. |

# Change Management

- **Request for Change** (RFC)
  - Documented request to alter a service or other Configuration Item (CI)
  - Can be submitted by anyone
    - e.g., Customers, IT staff, Users, etc.
  - Often Received by Service Desk and handled via the Change Management process

- The content of change management can be summarized as **7Rs**:
  - Who **raised** the change?
  - What is the **reason** for the change?
  - What is the **return** required from the change?
  - What are the **risks** involved in the change?
  - What **resources** are required to deliver the change?
  - Who is **responsible** for the build, test and implementation of the change?
  - What is the **relationship** between this change and other changes?

# Request for Change Form

RFC Number:

RFC Status:

**Status Options:** New, Awaiting Assessment, Assessed, Rejected by CM, Authorised, Rejected by CAB, In Build, In Test, Awaiting Implementation, Implemented, Awaiting Review, Closed

## To be completed by the Change Initiator:

Name:

Date RFC Raised:

Date Change is Required By:

System or Configuration Item to be changed:

Reason/Business Justification for the Change:

## To be completed by the Change Manager

Date Change Assessed by CM:

Priority Assigned:

**Priority Information**

**E Emergency, needs doing immediately (Emergency Change Process)**
**H High, needs doing within 48 hours**
**M M Medium, needs doing within 5 days**
**L Low, needs doing by indicated date**

Category Assigned:

**Category Information**

**Standard Change (via Heat) – Using a Procedure**
**IT Change Model – Using a Procedure**
**Minor Change – Authorised by Change Manager Alone**
**Significant Change – Authorised by a CAB**
**Major Change – Authorised by a CAB (Senior Level)**
**Emergency – Authorised by the ECAB**

If Rejected:
Reason for Rejection at Initial Assessment:
*(if appropriate)*

Date Change Initiator Informed of Change Rejection:
*(if appropriate)*

Risk of Change to the Business:
(H/M/L)

Impact of Change to the Business:
(H/M/L)

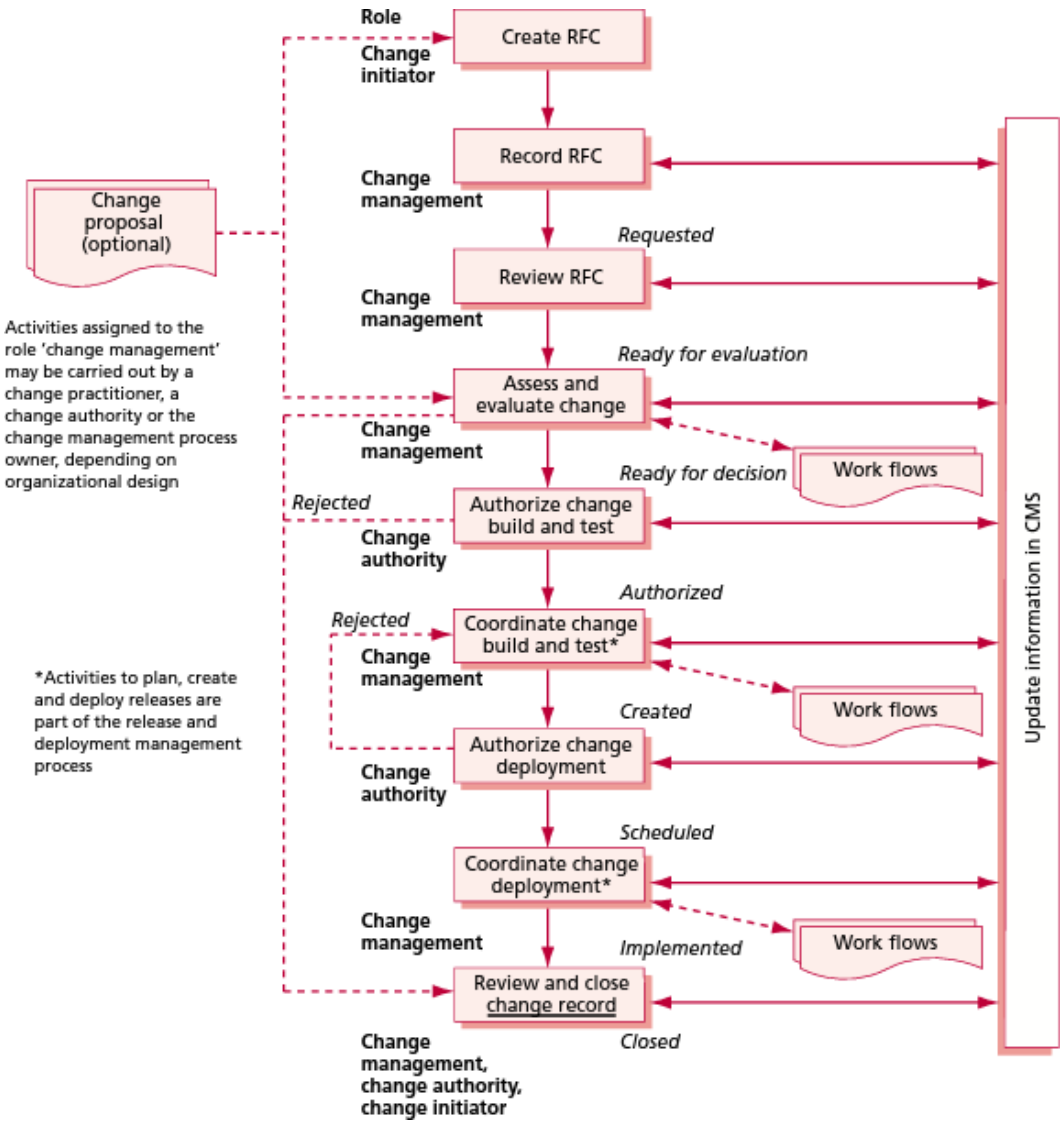
Change Authorised by (include all CAB members):

Does the Change Implementation Require Third Party/Supplier Involvement: **Y/N**

Name of the Third Party/Supplier:

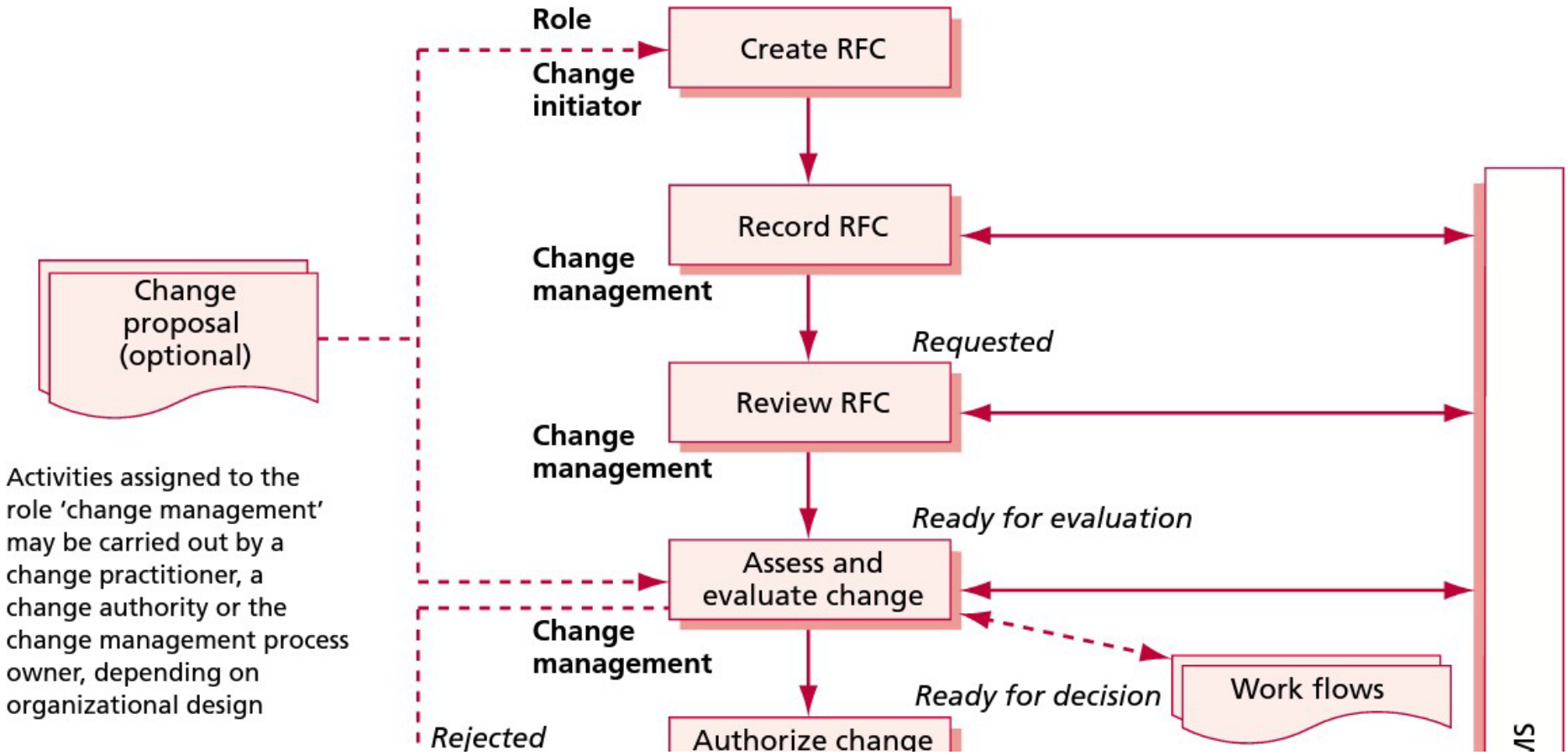
| Additional Information | Full Name | IT Team |
|---|---|---|
| Change Builder Identified | | |

# Change management, normal change request



# Change management, normal change request

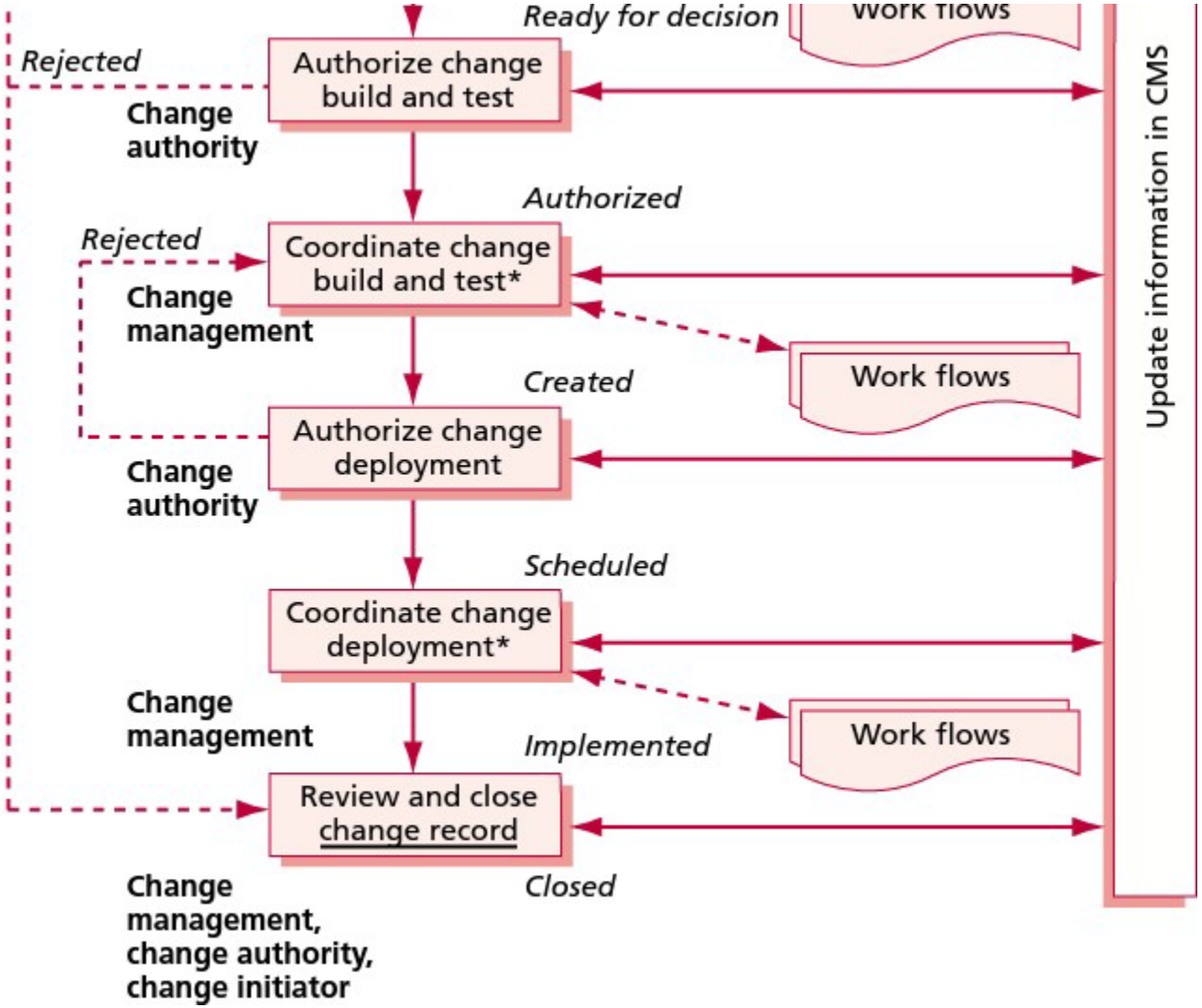




# Change management, normal change request



organizational design




*Activities to plan, create and deploy releases are part of the release and deployment management process

# Service Level Management

*The purpose of the **service level management** practice is to set clear business-based targets for service levels, and to ensure that delivery of services is properly assessed, monitored, and managed against those targets.*

A **service level** is *"one or more metrics that define expected or achieved service quality"*.

# Service Level Management

- SLM results in the creating of **Service Level Agreements (SLA**) between customers and the provider
- A service level agreement is ***"a documented agreement between a service provider and a customer that identifies both services required and the expected level of service"***.
- Service level agreements (SLAs) are used to measure the service performance from the customer's point of view

Related contracts/agreements:

- **Operational Level Agreements** (OLA) are performance agreements which exist between parts within the service provider organization
  - For supporting upstream SLAs
- **Underpinning contracts** (UC)
  - Agreement between service provider and third party

# Successful SLAs should

- Be related to a defined service so the scope is clear
- Relate to outcomes, not just operational metrics like ‘99% availability
- Reflect an agreement between the customer and service provider organisation
- Be simple to read and understand.

*Service level management provides end-to-end visibility of an organisation’s services:*

- It establishes a shared view of services and target service levels.
- It collects, analyses, stores and reports on relevant metrics.
- It performs service reviews and identifies improvement opportunities.
- It captures and reports on service issues.

# Service Level Management

## Examples of activities in the work

- Agree with customers
- Define service levels
- Document service levels
- Control and monitor (KPIs and metrics) including calculates achievements
- Change and improve agreements (SLA, OLA, UC)
- Reporting to mgmt. and customers
- Improve communication with business

# SLA exempel på innehåll

- SLA-partner:
  - Leverantör: [Leverantörsnamn]
  - Kund: [Kundnamn]
- Syfte:
  - Definiera tjänstenivåer för [Tjänstenamn].
  - Säkerställa gemensam förståelse av krav och ansvar.
- Tjänst:
  - [Kort beskrivning, t.ex. 'IT-support för interna system'].
- Omfattning:
  - [T.ex. 'Hantering av användarförfrågningar och incidenter kopplade till ERP-systemet'].

# SLA exempel på innehåll

- Mått och mål:
  - Incidentrespons: 90 % inom 1 timme.
  - Incidentlösning: 80 % inom 8 timmar.
  - Tillgänglighet: 99,5 % månatlig tillgänglighet.
- Mätmetod:
  - Automatiserade loggar från ärendehanteringssystemet.
- Kontorstid:
  - Måndag – Fredag, kl. 08:00 – 18:00.
- Kritiska incidenter:
  - 24/7 tillgång, svar inom 1 timme.

# SLA exempel på innehåll

- Leverantör:
  - Snabb och effektiv support enligt servicenivåer.
  - Informera om planerat underhåll 48 timmar i förväg.
- Kund:
  - Rapportera incidenter via [kanaler].
  - Tillhandahålla korrekt information för att påskynda lösningar.
- SLA gäller ej vid:
  - Force majeure (t.ex. naturkatastrofer, elavbrott).
  - Fel orsakade av kunden (t.ex. otillåten modifiering av systemet

# Översyn

- Granskning: Var 6:e månad.
- Syfte: Säkerställa att avtalet möter aktuella affärs- och kundbehov.

## Detta ingår i våra service- och driftavtal:

Serviceavtal
Enskilda enheter

Brons
8 timmar

Silver
4 timmar

Guld
2 timmar

# Serviceavtal

| Funktion | Brons | Silver | Guld |
|---|---|---|---|
| Prioriterad ärendehantering | Ja | Ja | Ja |
| Underhållsgenomgång av enheten * | Ja | Ja | Ja |
| Fjärrsupport till enheten | Ja | Ja | Ja |
| Rabatt på ord. taxa för konsulttjänster | Ja | Ja | Ja |
| Garanterad inställelsetid av tekniker ** | 8 tim | 4 tim | 2 tim |

*Serviceavtal kan även tecknas på tredje parts enheter som ACON inte har levererat efter genomgång.*

* Undehållsgenomgången görs utifrån utarbetade checklistor för er enhet och verksamhet och resulterar i en åtgärdslista med förslag på underhåll/förbättringar.

** Inställestetiden för tekniker gäller under kontorstid. Med kontorstid menas vardagar (måndag-fredag) kl 08.00 till 17.00, svensk tid. Undantaget är svenska helgdagar som infaller på vardagar samt dag före svenska helgdagar då kontorstid räknas från 08.00 till 12.00, svensk tid.

Downloaded from www.acon.se

# *Examples* of metrics in SLA

| Service level | Description |
|---|---|
| Service time / opening hours | The timeframe the IT service should be available |
| Capacity | Can be volume, number of transactions and number of users |
| Answering time / response time | For user support, it means how long it takes before the user reaches the technician. |
| Response time (on site) | Technician should begin the case within a certain time. |
| Repair time | The maximum time to restore the function. |
| Interruption frequency / Reliability | How often incidents may occur. |
| Availability | How high availability the service should have. Stated as a percentage, and means that the service is available to the customer. |

## SLA – calculation exercise

A customer buys SLA-package Multi in a Nordic Company.

- The customer perceived an incident at 07:38 at April the 2nd. It was immediately reported to the provider's service desk and the errand receives the errand id T456. At 11:30 o'clock the same day the provider calls and says that the service is up and running again.
- On May the 3rd the customer receives a report that indicate that there has been another incident between 22:11 and 04:00 o'clock on April the 14th (the timestamps are agreed with the customer).
- The operating time is all days from 00:00 to 24:00 (7/24). The 6-hour monthly maintenance window does not match the incidents.

a) Calculate the availability for the service during April.

b) Is the customer entitled to a penalty payment? If the answer is yes, how much should be repaid by the supplier?

**Basic information:**

| | | |
|---|---|---|
| Installation fee | 19000 SEK | Guaranteed availability= 99.2 % |
| Monthly fee (periodic recurrent fee) | 8900 SEK | Maintenance window = 6h |
| Contract time | 24 months | |

# a) Calculate the availability during April

Incident #1 (April the 2nd): 7:38 – 11:30 = 3:52

Incident #2 (April the 14th): 22:11 – 04:00 = 5:49

Summed up out-of-service time = 9:41 => 9.68 h

(Note! 41m/60m=0,68)

Operation time during April = 30 days * 24 h = 720 h

Maintenance window 6 h per month

Break: 9.68 / (720 – 6) = 0.01356 = 1.36 %

Availability = 100 – 1.36 % = 98.64 %

Alternative:

$$\frac{(720 - 6 - 9.68)}{(720 - 6)} =$$

98.64 %

## SLA – calculation exercise

b) Is the customer entitled to a penalty payment? If the answer is yes, how much should be repaid by the supplier?

Guaranteed availability = 99.2 %

Actual availability = 98.64 %

- Since the actual availability is below the guaranteed level the customer is entitled to a penalty payment. How much?
- We have to find out how many hundredths percent points the actual availability is less than the guaranteed availability:

99.20 – 98.64 = 0.56 (i. e. 56 hundredths percent points)

Note! For each hundredth that differs from agreed level of availabiliy the customer is entitled to 0.25 % of the monthly fee.

Summed up the customer is entitled to 56 * 0.25 % = 14 %

Monthly fee = 8900 SEK

Repayment = 14 % of 8900 = 0.14 * 8900 = 1246 SEK

# What are the benefits of an SLA?

FOR SERVICE PROVIDERS & SERVICE CUSTOMERS?

# Possible benefits

- For service provider/supplier:
  - Manage expectations
  - Be able to ”control the customer”
  - IT knows what to deliver, when, to who and how
  - Promotes proactivity and planning for investments
  - Improves customer relationship?
- For service customer/client
  - Manage expectations
  - Balance between cost and quality
  - Understand what is included and what is not included
  - Improved supplier relationship
  - Know what you get for the money